

CRISPR/cas9 cassette targeting *Escherichia coli* ^{bla}CTX-M specific gene of mastitis cow milk origin can alter the antibiotic resistant phenotype for cefotaxime

Rajib Deb, Parul Chaudhary & Sachinandan De

To cite this article: Rajib Deb, Parul Chaudhary & Sachinandan De (2022): CRISPR/cas9 cassette targeting *Escherichia coli* ^{bla}CTX-M specific gene of mastitis cow milk origin can alter the antibiotic resistant phenotype for cefotaxime, Animal Biotechnology, DOI: [10.1080/10495398.2022.2053695](https://doi.org/10.1080/10495398.2022.2053695)

To link to this article: <https://doi.org/10.1080/10495398.2022.2053695>



Published online: 31 Mar 2022.



Submit your article to this journal [↗](#)



Article views: 143



View related articles [↗](#)



View Crossmark data [↗](#)



CRISPR/cas9 cassette targeting *Escherichia coli* ^{bla}CTX-M specific gene of mastitis cow milk origin can alter the antibiotic resistant phenotype for cefotaxime

Rajib Deb^{a,b}, Parul Chaudhary^a, and Sachinandan De^a

^aAnimal Genomics Laboratory, Animal Biotechnology Center, ICAR-National Dairy Research Institute, Karnal, India; ^bICAR-National Research Center on Pig, Guwahati, Assam, India

ABSTRACT

CTX-M beta-lactamases are one of the most important extended spectrum beta-lactamase (ESBL) resistance enzymes found in *E. coli*. In the present study, 59% of *E. coli* isolates from mastitis cow milk were reported to be positive for ESBL types. The prevalence of beta-lactam (β -lactam) antibiotic resistance was reported to be 84%, 72.7%, 52.27%, 50%, and 45.4% for cefotaxime, cefepime, cefuroxime, oxacillin, and cephalexine, respectively. The ^{bla}CTX-M gene was found in 65% ($n = 17$) of the *E. coli* isolates when they were genotyped. Further, the use of a CRISPR/cas9 cassette to target the *E. coli* ^{bla}CTX-M gene revealed changes in antibiotic phenotypes for cefotaxime.

KEYWORDS

CRISPR/cas9; CTX-M beta-lactamases; mastitis; *E. coli*; ^{bla}CTX-M gene

Introduction

Bovine mastitis is one of the most costly and complex diseases to treat in the dairy industry. It has a deleterious influence on animal health and production, as well as posing a risk to consumers' health. Bovine mastitis is a complex illness that arises from the combination of different variables related to the host, particular infections, the environment, the season, and farm management.¹ Although other bacterial species have been recovered from bovine mastitis cases, *Staphylococci* and *Escherichia coli* (*E. coli*), as well as other members of the Enterobacteriaceae family, have historically been the most prevalent mastitis causes.^{2–4}

In veterinary medicine, beta-lactam (β -lactam) antibiotics are still the first-line treatment, and their widespread use in the prevention and treatment of mastitis poses a risk of antibiotic residues in milk and the development of multidrug-resistant bacteria, which could be transmitted to human consumers via raw milk and its dilution.⁵ The widespread usage of β -lactam antimicrobials has resulted in the establishment of resistant bacteria all over the world. Acquisition of β -lactamase genes found on mobile genetic elements such as plasmids or transposons is the most common mechanism for β -lactamase resistance.⁶ The majority of β -lactamases discovered in *E. coli* belong to Ambler class A, which is further split into narrow-spectrum β -lactamases

(TEM-1, TEM-2, and SHV-1) and extended-spectrum β -lactamases (ESBLs) (TEM-3, SHV-5, and CTX-M-like).⁷ Extended-spectrum cephalosporins, which are often used to treat *E. coli* infections, are resistant to ESBLs. CTX-M-type β -lactamases (CTX-Ms) are broad-spectrum β -lactamases that are formed from chromosomally encoded β -lactamases of *Kluyvera* species.⁸

CRISPR-Cas9 was discovered in bacterial immune systems and has since been engineered to create site-specific double-strand breaks (DSBs) for genome editing in a variety of organisms, including mammalian cells, plants, fungi, yeast, and bacteria.⁹ A Cas9 nuclease from *Streptococcus pyogenes* for DNA cleavage and an artificial chimeric single guide RNA (sgRNA) for target recognition make up the designed CRISPR-Cas9 system.¹⁰ Few studies highlighted the application of CRISPR/cas9 toward reversal of various antibiotic resistant phenotypes in AMR bacteria.^{11–14} In this preliminary study, we develop a CRISPR/Cas9 construct targeting *E. coli* ^{bla}CTX-M specific gene originated from mastitis cow milk samples.

Materials and methods

Mastitis milk samples and enrichment

A total of 44 milk samples were collected from the Institute Cattle Yard, NDRI, Karnal, India, from

whole tits of mastitis cows. Samples were supplemented in sterile Luria Bertani (LB) Broth (HIMEDIA, India) and incubated overnight at 37 °C in an incubator-shaker, then active cultures were streaked on HiCrome™ ESBL agar base (HIMEDIA, India), and the resultant bacterial plates were incubated at 37 °C for 12–16 h. Bacterial colonies on HiCrome™ ESBL agar base plates with pink to purple colonies and typical *E. coli* shape as well as lactose fermentation capabilities were chosen and stored at –80 °C in Luria-Bertani (LB) broth supplemented with 100% glycerol.

Antibiotic sensitivity testing

All positive *E. coli* isolates from mastitis cow milk were tested for antibiotic susceptibility with various β -lactam antibiotic classes (oxacillin, cephalaxine, cefuroxime, cefotaxime, cefepime and meropenem). The isolates' antimicrobial susceptibility was tested using the conventional Kirby–Bauer disk diffusion technique on Mueller–Hinton agar medium (Merck, Germany) using commercially available or laboratory generated 5–6 mm antibiotic disks. For each antibiotic disk, the diameter of the inhibition zone was measured, and the findings were characterized in accordance with the CLSI recommendations.¹⁵

CTX-M genotypic characterization

Using Chen and Kuo's¹⁶ approach with minor modifications, genomic DNA was recovered from an overnight growth *E. coli* culture isolated from milk samples of mastitis-affected cows. In a nutshell, enriched *E. coli* cells from milk samples were extracted at the conclusion of their exponential phase of development by centrifugation for 5 min at 12,000 rpm. Cell pellets were resuspended in lysis buffer (40 mM Tris-acetate pH 7.8, 20 mM sodium-acetate, 1 mM EDTA, and 1% SDS), vigorously pipetted, and centrifuged at 12,000 rpm for 10 min at 4 °C. The supernatant was transferred to a new vial, which was then filled with an equal volume of chloroform and inverted many times until the solution appeared milky, then the mixture was centrifuged at 12,000 rpm for 3 min and the supernatant was shifted to another tube for DNA precipitation with 100% ethanol, washed twice with 70% ethanol, air-dried, and resuspended in 1 × TE buffer. Spectrophotometric data were used to assess the purity of the isolated genomic DNA.

Table 1. Synthesized oligos for the present study.

Sl. no.	Oligos	Sequence
1	^{bla} CTX-M Forward	CGCGGTGCTGAAGAAAAGTG
2	^{bla} CTX-M Reverse	GCCGGTTTTATCCCCACAA
3	SgRNA Forward	AAACCCGTGCGCATCAGCGTGAACG
4	SgRNA Reverse	GGCAGCGGTAGTCGCACTTGCAAAA
5	Flanking primer	TTGATTTGAGTCAGCTAGGAGG

Extracted genomic DNA was treated to polymerase chain reaction (PCR) with ^{bla}CTX-M gene-specific primers (Table 1, Sl. no. 1 and 2). 10 × PCR buffer (50 mM KCl, 10 mM Tris-HCl, 0.1 percent Triton X), 5 U Taq DNA Polymerases, 1 mM MgCl₂, 1.5 M each primer, and 200 M each dNTP were used in a 25 µl reaction volume. PCR was carried out for 40 cycles of amplification, with denaturation at 95 °C for 5 minutes, annealing at various temperatures (55–60 °C, 72 °C for 15 sec), and extension at 72 °C for 5 min. The PCR results were verified by electrophoresis on an agarose gel.

Construction of CRISPR/cas9 cassette targeting *E. coli* CTX-M specific gene

CRISPOR tool¹⁷ was used to create guide RNA oligos targeting a conserved area of the ESBL ^{bla}CTX-M gene, and the oligo pairs were synthesized (Table 1, Sl. no. 3 and 4). Luciano Marraffini (Addgene plasmid # 42876; <http://n2t.net/addgene:42876>; RRID:Addgene 42876) provided the pCas9 plasmid vector utilized in this investigation as a gift. *Bsa*I restriction enzyme (NEB, USA) was used to digest the vector, which was then purified using the Qiagen gel extraction kit (USA). T4 polynucleotide kinase was used to phosphorylate both oligos (NEB, USA). 2.5 µl of 1 M NaCl was added to the phosphorylated oligo pairs and incubated at 95 °C for 5 min before cooling to room temperature in the thermocycler. T4 ligase was used to ligate gel purified, *Bsa*I digested pCas9 with annealed oligos diluted 10 times (NEB, USA). For 2 h, the ligated mixture was incubated at room temperature. In *E. coli* DH5 cells, the ligated product was transformed. Colony PCR using reverse oligo and flanking sequence primers was used to validate positive clones (Table 1, Sl. no. 4 and 5). Sequencing was used to confirm the positive clones.

In vitro assessment

Gel purified recombinant pCas9 plasmid was transformed in different ($n=7$) *E. coli* CTX-M cells isolated from the clinical mastitis cow milk samples using electroporation method described by Gonzales et al.¹⁸ Transformed bacterial cells were subjected for

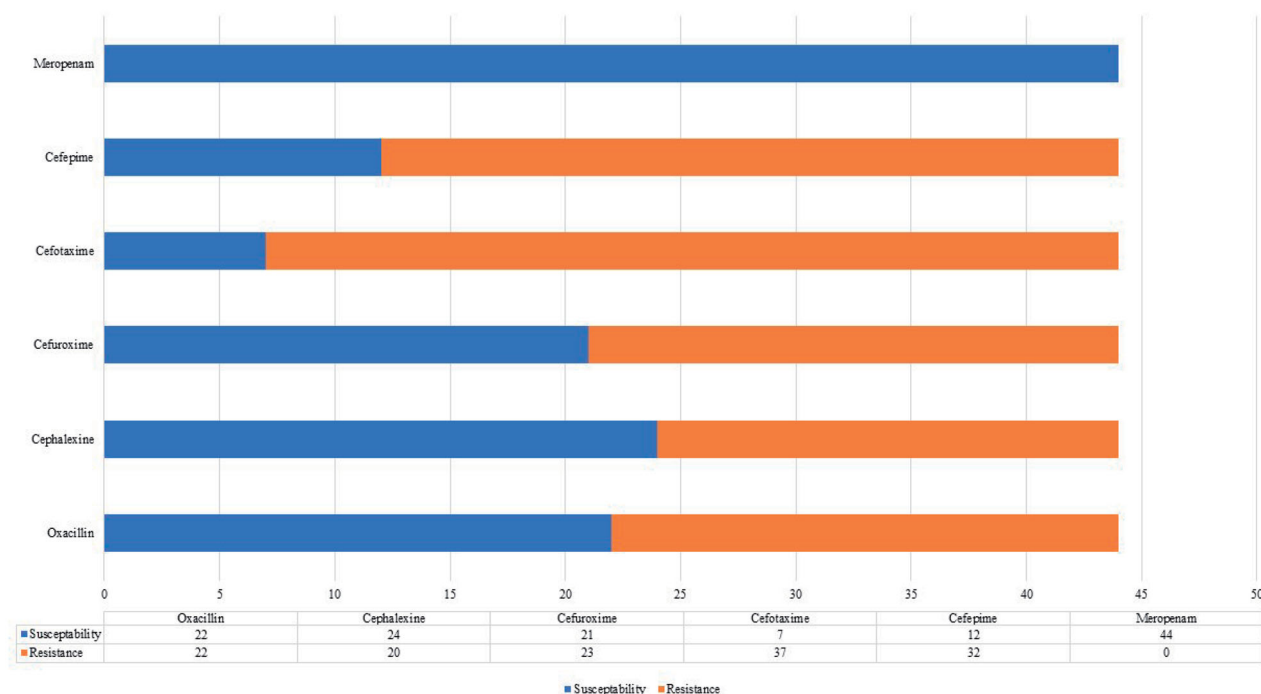


Figure 1. Different β -lactam antibiotics susceptibility pattern of the isolated ESBL *E. coli* from mastitis cow milk.

antibiotic sensitivity testing for different β -lactam antibiotics as mentioned earlier.

Results and discussion

Present study was designed to construct and characterize a CRISPR/cas9 cassette targeting *E. coli* bla_{CTX-M} specific gene originated from mastitis cow milk. Out of 44 samples, 59% ($n=26$) *E. coli* isolates of mastitis milk origin was found to be positive for ESBL types. The prevalence of different β -lactam antibiotic resistance was found to be 84%, 72.7%, 52.27%, 50%, 45.4% for cefotaxime, cefepime, cefuroxime, oxacillin and cephalexine, respectively (Fig. 1). However, we could not get any resistant phenotypes for meropenem antibiotics. When the *E. coli* isolates were genotyped, it was revealed that 65% ($n=17$) of them had bla_{CTX-M} gene (Fig. 2). Antibiotic susceptibility profiles of *E. coli* ($n=40$) isolated from postpartum uterine infection in dairy cattle in India revealed significant rates of resistance to ampicillin (95.0%), cefpodoxime (97.5%), cefotaxime (87.5%), ceftriaxone (87.5%), ceftazidime (25%), amoxicillin/clavulanic acid (20%), cefazidime (17.5%), gentamicin (10%), and ertapenem (10%) according to Agrawal et al.¹⁹ A total of 65 *E. coli* isolates with the ESBL phenotype were identified in veterinary clinical samples, with the bla_{CTX-M1} and bla_{CTX-M2} genes being the most prevalent.²⁰ bla_{CTX-M} genes were detected in 183 isolates (93.4%) of ESBL-producing *E. coli*-positive bacteria from dairy

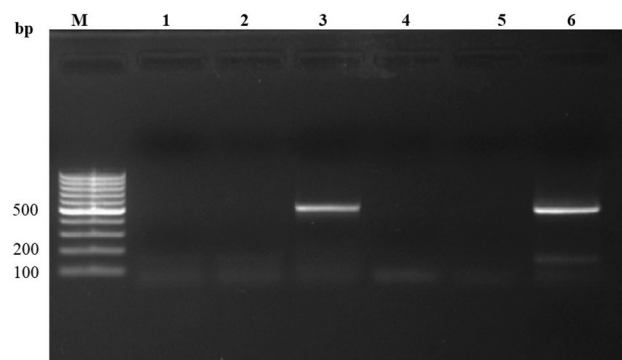


Figure 2. Screening of bla_{CTX-M} gene from the ESBL *E. coli* isolates from mastitis cow milk samples. Lanes 1–2, 4–5: Negative for bla_{CTX-M} gene; Lanes 3 and 6: Positive for bla_{CTX-M} gene showing 475 bp DNA fragment.

farms in the southern region of Bavaria, Germany, with CTX-M group 1 being the most often found group.²¹ ESBL-producing strains were found in 20% of *E. coli* isolates from beef samples in Jimma town, Southwest Ethiopia, and 85.7% of ESBL-producing germs were resistant to cefotaxime and ceftriaxone, and 71.4 percent to ceftazidime.²² The bla_{CTX-M} gene was found to be the most common ESBL gene among *E. coli* strains, accounting for 21 (9.5%) of all isolates, followed by bla_{SHV} , which was discovered in 18 (8.1 percent) of isolates from beef cattle in China's Sichuan-Chongqing Circle.²³ bla_{TEM} , bla_{SHV} , and bla_{CTX-M} genes were found in *E. coli* isolates from cattle farms and raw beef in the North-West province

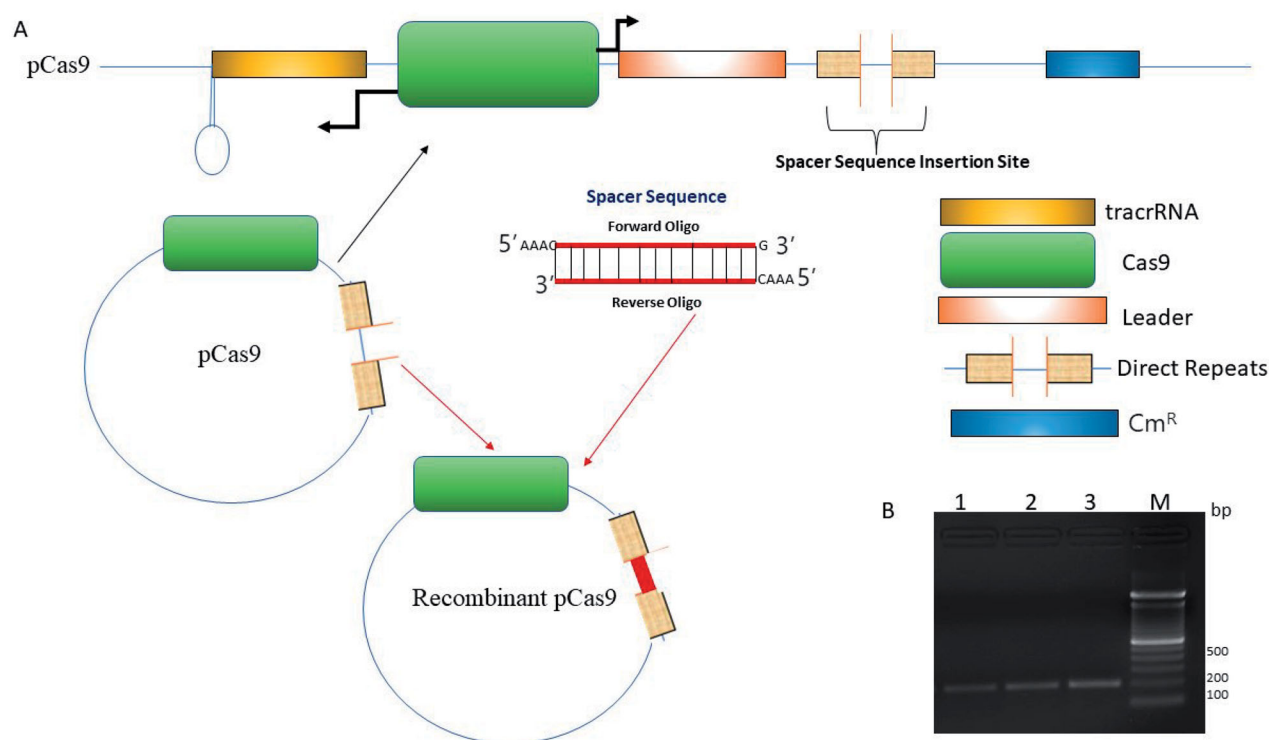


Figure 3. Construction of recombinant pCas9 plasmid encoding spacer sequence (annealed guide RNA oligos). (A) Schematic representation on the preparation of the construct. pCas9 plasmid consist of tracrRNA (Trans-activating crRNA), cas9 endonuclease, leader sequence, direct repeat having the spacer sequence insertion site and chloramphenicol resistant genes (Cm^R). Spacer sequence was prepared by annealing of phosphorylated guide RNA oligos having overhang sequences. (B) Recombinant plasmids were confirmed by colony PCR using reverse oligo primer and a designed flanking sequence forward primer.

of South Africa was 85.5%, 69.6%, and 58%, respectively.²⁴ Out of 153 *E. coli* isolates from mastitic milk samples ($n = 1252$) from China, 36 (23.53%) were verified as ESBL-producers, and nucleotide analysis of PCR amplicons indicated that bla_{CTX-M} was the major ESBL gene identified in 28 (77.78%) isolates.²⁵ Similarly, Feng et al.²⁶ identified the ESBL prevalence (22.96%) in different commercial dairy herds of China and observed that, the prevalence of bla_{CTX-M} and bla_{TEM-1} was 97.26% and 71.23%, respectively.

To develop a CRISPR-cas system for altering the antibiotic resistant phenotypes of ESBL *E. coli*, we designed and synthesized a set of guide RNA oligos. Phosphorylated annealed oligos were cloned in a prokaryotic expression system encoding cas9 endonuclease. Positive clones were confirmed by colony PCR depicting the amplified product of 210 bp. Recombinant plasmid construct was transformed in the competent ESBL *E. coli* cells bearing bla_{CTX-M} gene (Fig. 3). Successfully transformed *E. coli* cells were subjected for antibiotic sensitivity assay using β -lactam antibiotics (oxacillin, cephalexine, cefuroxime, cefotaxime, cefepime and meropenem). Results revealed that, transformed cells were susceptible cefotaxime in comparison to the non-transformed *E. coli*

CTX-M cells (Fig. 4). Kim et al.¹² used the CRISPR-cas system to re-sensitize antibiotic-resistant *E. coli* with TEM- and SHV-type ESBLs. CRISPR-cas9 insertion into an *E. coli* model strain carrying the bla_{TEM1} gene resulted in a complete phenotypic reversal to an antibiotic-sensitive phenotype.¹³ CRISPR/cas9 genome editing was also used by Wu et al.¹⁴ to reverse carbapenem resistance in *Shewanella algae*. de Maat et al.¹¹ again reported the CRISPR-Cas9-mediated genome editing in vancomycin-resistant *Enterococcus faecium*.

Although antimicrobial peptides (AMPs) and phage therapy are promising tools for combating antimicrobial resistance, they have a number of drawbacks, including a limited bacterial host range (strict host strain specificity), the need for multiple phages to control multiple bacterial pathogens, and knowledge of the infectious agent; high production costs; safety concerns; the risk of resistance; and poor transport stability.^{27,28} In this connection, CRISPR system may take advantages. CRISPR/Cas9 technology has been used to create sequence-specific antimicrobials that only target antibiotic-resistant bacteria, according to studies.^{29,30} The findings show that programmed Cas9 mediates the cytotoxicity of resistant cells by targeting specific sequences. These findings suggest that



Figure 4. β -Lactam antibiotic phenotypes of ESBL *E. coli* bearing CTX-M genes. (A) Without transforming the recombinant pCas9 plasmid; (B) After transforming with the recombinant pCas9 plasmid.

antibiotic-resistant bacteria could be induced to revert to antibiotic-sensitive cells by using the CRISPR/Cas9 system to specifically cleave resistance genes.

This work demonstrates how the CRISPR/cas9 system may be used to provide a unique strategy to combat antibiotic resistance in dairy cattle, as well as ways to mitigate it. However, developing appropriate strategies for in vivo delivery of CRISPR cassettes in animals will undoubtedly pave the way for the development of appropriate antimicrobial resistance mitigation strategies in dairy animals.

Acknowledgements

The authors are thankful to Director, ICAR-National Dairy Research Institute, Karnal, Haryana, India for providing the necessary facilities to conduct the experiment.

Disclosure statement

We declare that none of the authors have any conflict of interest for the present study.

Funding

The authors acknowledge the financial support received from the Indian Council of Agricultural Research, New Delhi, India as ICAR-Post Doctoral Fellowship to RD during this study period. The authors are also thankful to Science and Engineering Research Board, Department of Science and Technology, Government of India for the financial assistant to SD for the ongoing project code [EMR/2017/004602].

References

1. Zhang Z, Li XP, Yang F, et al. Influences of season, parity, lactation, udder area, milk yield, and clinical symptoms on intramammary infection in dairy cows. *J Dairy Sci.* 2016;99(8):6484–6493.
2. Hogan J, Smith KL. Coliform mastitis. *Vet Res.* 2003; 34(5):507–519.
3. Piessens V, Van Coillie E, Verbist B, et al. Distribution of coagulase-negative *Staphylococcus* species from milk and environment of dairy cows differs between herds. *J Dairy Sci.* 2011;94(6):2933–2944.
4. Taponen S, Pyörälä S. Coagulase-negative staphylococci as cause of bovine mastitis- not so different from *Staphylococcus aureus*? *Vet Microbiol.* 2009; 134(1–2):29–36.
5. Klibi A, Jouini A, Boubaker El Andolsi R, et al. Epidemiology of β -lactamase-producing *Staphylococci* and gram negative bacteria as cause of clinical bovine mastitis in Tunisia. *Biomed Res Int.* 2019;2019: 2165316.
6. Liu B, Yu K, Ahmed I, et al. Key factors driving the fate of antibiotic resistance genes and controlling strategies during aerobic composting of animal manure: a review. *Sci Total Environ.* 2021;791:148372.
7. Paterson DL, Bonomo RA. Extended-spectrum beta-lactamases: a clinical update. *Clin Microbiol Rev.* 2005; 18(4):657–686.
8. Cantón R, González-Alba JM, Galán JC. CTX-M enzymes: origin and diffusion. *Front Microbiol.* 2012; 3:110.
9. Lino CA, Harper JC, Carney JP, Timlin JA. Delivering CRISPR: a review of the challenges and approaches. *Drug Deliv.* 2018;25(1):1234–1257.
10. Jinek M, Chylinski K, Fonfara I, Hauer M, Doudna JA, Charpentier E. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science.* 2012;337(6096):816–821.

11. de Maat V, Stege PB, Dedden M, et al. CRISPR-Cas9-mediated genome editing in vancomycin-resistant *Enterococcus faecium*. *FEMS Microbiol Lett*. 2019; 366(22):p.fnz256.
12. Kim JS, Cho DH, Park M, et al. CRISPR/Cas9-mediated re-sensitization of antibiotic-resistant *Escherichia coli* harboring extended-spectrum β -lactamases. *J Microbiol Biotechnol*. 2016;26(2):394–401.
13. Tagliaferri TL, Guimarães NR, Pereira MDP, et al. Exploring the potential of CRISPR-Cas9 under challenging conditions: facing high-copy plasmids and counteracting beta-lactam resistance in clinical strains of Enterobacteriaceae. *Front Microbiol*. 2020;11:578.
14. Wu ZY, Huang YT, Chao WC, Ho SP, Cheng JF, Liu PY. Reversal of carbapenem-resistance in *Shewanella algae* by CRISPR/Cas9 genome editing. *J Adv Res*. 2019;18:61–69.
15. CLSI W. 2018. *Clinical and Laboratory Standards Institute: Performance Standards for Antimicrobial Susceptibility Testing*. 28th ed. CLSI Supplement M100.
16. Chen WP, Kuo TT. A simple and rapid method for the preparation of gram-negative bacterial genomic DNA. *Nucleic Acids Res*. 1993;21(9):2260.
17. Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. *Nucleic Acids Res*. 2018;46(W1): W242–W245.
18. Gonzales MF, Brooks T, Pukatzki SU, Provenzano D. Rapid protocol for preparation of electrocompetent *Escherichia coli* and *Vibrio cholerae*. *J Vis Exper*. 2013; 8(80):50684.
19. Agrawal S, Singh AP, Singh R, et al. Molecular characterization of extended-spectrum β -lactamase-producing *Escherichia coli* isolated from postpartum uterine infection in dairy cattle in India. *Vet World*. 2021;14(1):200–209.
20. Huang YH, Kuan NL, Yeh KS. Characteristics of extended-spectrum β -lactamase-producing *Escherichia coli* from dogs and cats admitted to a veterinary teaching hospital in Taipei, Taiwan from 2014 to 2017. *Front Vet Sci*. 2020;7:395.
21. Schmid A, Hörmansdorfer S, Messelhäuser U, Käsbohrer A, Sauter-Louis C, Mansfeld R. Prevalence of extended-spectrum β -lactamase-producing *Escherichia coli* on Bavarian dairy and beef cattle farms. *Appl Environ Microbiol*. 2013;79(9):3027–3032.
22. Abayneh M, Tesfaw G, Woldemichael K, Yohannis M, Abdissa A. Assessment of extended-spectrum β -lactamase (ESBLs)-producing *Escherichia coli* from minced meat of cattle and swab samples and hygienic status of meat retailer shops in Jimma town, Southwest Ethiopia. *BMC Infect Dis*. 2019;19(1):1–8.
23. Zhang YL, Huang FY, Gan LL, et al. High prevalence of blaCTX-M and blaSHV among ESBL producing *E. coli* isolates from beef cattle in China's Sichuan-Chongqing Circle. *Sci Rep*. 2021;11(1): 13725.
24. Montso KP, Dlamini SB, Kumar A, Ateba CN. Antimicrobial resistance factors of extended-spectrum beta-lactamases producing *Escherichia coli* and *Klebsiella pneumoniae* isolated from Cattle Farms and Raw Beef in North-West Province, South Africa. *Biomed Res Int*. 2019;2019:4318306.
25. Ali T, Ur Rahman S, Zhang L, et al. ESBL-producing *Escherichia coli* from cows suffering mastitis in China contain clinical class 1 integrons with CTX-M linked to ISCR1. *Front Microbiol*. 2016;7:1931.
26. Yang F, Zhang S-d, Shang X-f, et al. Prevalence and characteristics of extended spectrum β -lactamase-producing *Escherichia coli* from bovine mastitis cases in China. *Journal of Integrative Agriculture*. 2018;17(6): 1246–1251. doi:10.1016/S2095-3119(17)61830-6.
27. Abedon ST, Loc-Carrillo C. Pros and cons of phage therapy. *Bacteriophage*. 2011;1(2):111–114.
28. Chen CH, Lu TK. Development and challenges of antimicrobial peptides for therapeutic applications. *Antibiotics*. 2020;9(1):24.
29. Bikard D, Euler CW, Jiang W, et al. Exploiting CRISPR-Cas nucleases to produce sequence-specific antimicrobials. *Nat Biotechnol*. 2014;32(11):1146–1150.
30. Citorik RJ, Mimee M, Lu TK. Sequence-specific antimicrobials using efficiently delivered RNA-guided nucleases. *Nat Biotechnol*. 2014;32(11):1141–1145.